# TerraTrace: Reproducible, Uncertainty-Aware, Multi-Lens Morphological Analog Screening Across Five Worlds

**NASA Space Apps 2026 - research preview.** Repository: `github.com/pal-123456789/terratrace`

> **Scope / claim firewall.** TerraTrace screens terrain *morphology* between curated map products. It does **not** estimate rover traversability, validate autonomy, establish geological equivalence, certify site access, or certify mission readiness. Earth sites named here are *illustrative analogs*, not endorsed field sites. It is not NASA-endorsed.


## Abstract

Choosing a terrestrial analog for a planetary terrain is usually done by expert judgment and reported as a single qualitative verdict. TerraTrace makes that screening **quantitative, reproducible, and honest**. It brings a planetary reference terrain and candidate Earth terrains to a common metric support on each body's own geometry, computes a battery of map-scale morphology descriptors, and ranks candidates by **Pareto dominance across several independent descriptor "lenses"** - deliberately refusing to collapse the comparison into one composite "likeness" score. When lenses disagree it returns an explicit TIE / INDETERMINATE state instead of inventing a winner, and attaches a spatial block-bootstrap, a permutation significance test, a non-composite decision-confidence verdict, and a full provenance record to every result. On a real five-world run (Mars MOLA vs three Copernicus GLO-30 Earth terrains) the method independently re-identifies a documented basalt analog - Craters of the Moon, Idaho - as the closest match on the core roughness/relief axes, while honestly reporting that the recommendation does not survive every lens. Everything is offline-reproducible from one command and covered by 686 automated tests.


## 1. Problem and approach

Analog site selection underpins instrument testing, operational rehearsal, and the interpretation of orbital data. Yet "X is a good analog for Y" is rarely stated with a defined support scale, an uncertainty, or a falsifiable procedure. TerraTrace treats analog screening as a transparent, auditable computation with three commitments: (i) **comparison at a declared common support** on each body's correct geometry; (ii) **no composite score** - candidates are ranked by Pareto dominance over interpretable axes, and ties are reported, not broken; and (iii) **every number carries provenance and uncertainty**. The goal is not to declare a winner but to say, defensibly, *what transfers, at what scale, and with what confidence.*

## 2. Method

**2.1 Common support and per-body reprojection.** Each area of interest is warped to its own local azimuthal-equidistant metric grid using the correct body radius (Mars 3,396,190 m; Moon 1,737,400 m; Mercury 2,439,400 m; Ceres 469,730 m; Earth WGS84) and aggregated to a declared common support (e.g. 1000 m), so descriptors compare like with like rather than conflating pixel size with terrain.

**2.2 Morphology descriptors.** Core: slope, local relief, detrended ruggedness, low-slope connectivity. Extended: aspect dispersion, hypsometric integral. Geomorphic: Zevenbergen-Thorne curvature and the Vector Ruggedness Measure. Landform: geomorphon landform composition and topographic openness. Scale & fabric: the radially-averaged power-spectral slope and a directional-structure-function anisotropy (fabric strength and orientation). Planetary roughness: the Kreslavsky-Head (2000) differential-slope baseline roughness (the metric behind NASA's global MOLA roughness maps) and the Iwahashi-Pike (2007) three-signature (slope / convexity / texture) nested-means terrain class. Dissection: a depression-filled D8 flow-accumulation surface-dissection texture (a purely geometric routing descriptor). The bank also carries a Haar wavelet multiscale-energy spectrum, the Strahler hypsometric curve, surface-normal-dispersion roughness, and a Weiss (2001) TPI slope-position classification. A Topographic Wetness Index (Beven & Kirkby 1979) and a convergence index (Kothe & Lehmeier 1996) summarise flow-geometric convergence; a sky-view factor (Zaksek et al. 2011) captures hemispherical openness; and a two-scale TPI slope-position classification (Weiss 2001) resolves nested landform position. Finally a differential box-counting fractal dimension, gliding-box lacunarity, Shannon terrain entropy, and a Flint (1974) slope-vs-contributing-area concavity complete the battery. A geostatistical semivariogram (range/sill/nugget; Matheron 1963) and an RMS/Allan-deviation roughness length (Shepard et al. 2001) summarise spatial structure; Wood (1996)

morphometric curvature classes and a Woodcock (1977) surface-normal orientation-tensor fabric add shape descriptors; an openness-difference slope position (Yokoyama 2002) and the Melton (1965) ruggedness number complete the batch. Distributions are compared to the reference with scale-normalized Wasserstein/KS distances.

**2.3 Decision: Pareto, not a score.** Candidates become vectors of distances-to-reference and are ranked by **Pareto dominance** (peeled non-dominated layers + a dominance matrix). The result is one of `RECOMMEND_FOLLOW_UP`, `TIE`, `INDETERMINATE`, `NOT_COMPARABLE`, or `BLOCKED`. Six **lenses** - core, extended, geomorphic, landform, texture, hydro - are independent axis sets (the hydro lens adds a Topographic Wetness Index, aspect convergence, and flow-dissection distance-to-reference axes); a recommendation that holds under more lenses is more robust, and lens disagreement is surfaced, not hidden.

**2.4 Uncertainty, significance, confidence.** A spatial moving-block bootstrap gives autocorrelation-aware distance intervals and a non-domination frequency; a one-sided permutation test asks whether the recommended unit is genuinely closer to the reference than the runner-up; and a **non-composite decision-confidence** verdict (`robust` / `qualified` / `fragile` / …) restates the bootstrap, significance, and lens-stability evidence as three independent signals, explicitly naming any signal it could not evaluate.

**2.5 Provenance and reproducibility.** Every study is hash-anchored (sha256), carries an evidence graph and a transfer-gap ledger, and exports a self-contained bundle (study package, Markdown + offline-HTML report, viewer, references, checksums, confidence). A fixed synthetic study is byte-reproducible within an environment; portable invariants (not pinned digests) gate it across platforms. The full geomorphometric descriptor bank and a swath/profile `transect` extractor are reachable from the unified CLI (`bank`, `transect`) and the loopback-only HTTP API (`POST /bank`, `POST /transect`, the same host-allowlist and body-cap guards as the existing endpoints), and the terrain fingerprint carries an opt-in geostatistical similarity term (`include_geostat`).

## 3. Data

Mars reference: MGS **MOLA 463 m** (USGS Astrogeology, keyless). Earth candidates: **Copernicus GLO-30** via OpenTopography. The engine's source catalog carries 13 products across the five bodies with coverage, licence, and 1-sigma vertical-accuracy metadata; a no-network preflight validates each bounding box, selects the covering product, and sizes the request before any download.

## 4. Result: a real five-world screen

Mars MOLA versus three real Copernicus GLO-30 Earth terrains at 1000 m support. Distance to the Mars reference per axis (lower = closer):

| candidate | slope | relief | ruggedness | connectivity | aspect | hypsometric |
|---|---|---|---|---|---|---|
| **Craters of the Moon, ID** | **0.75** | **0.78** | **0.76** | **0.018** | 0.39 | 0.28 |
| **Nevada basin** | 1.33 | 1.38 | 1.45 | 0.030 | **0.011** | **0.14** |
| Andean volcano | 1.73 | 1.50 | 1.85 | 0.057 | 0.012 | 0.17 |

Four-lens ablation on the **same** data (one command, `--ablation`):

| lens | result | recommended | non-dominated set |
|---|---|---|---|
| core | RECOMMEND | **Craters of the Moon** | {Craters} |
| extended | TIE | - | {Craters, Nevada} |
| geomorphic | RECOMMEND | **Craters of the Moon** | {Craters} |
| landform | TIE | - | {Craters, Nevada, Andes} |

**Reading.** On raw roughness/relief/ruggedness (core) and on curvature/vector-ruggedness (geomorphic), Craters of the Moon is decisively the closest match - the method *independently re-identifies a terrain already used as a basalt analog.* Adding aspect-dispersion and hypsometric structure (extended), where the Nevada basin matches Mars better and even shares its ~45 deg fabric, turns the decision into an honest TIE; the landform lens cannot separate the three. The defensible conclusion is therefore **lens-conditional**: *Craters of the Moon is the robust*

*analog for surface roughness and local shape; it is not uniquely best for directional/elevation structure.* No single number could have said that honestly.

**Corroboration at scale.** Running the one-command atlas over four distinct Mars terrains (Jezero and Gale crater floors; Daedalia Planum and Elysium Planitia volcanic plains) against the same three Earth candidates under all four lenses (16 referencexlens cells) makes the result sharper: Craters of the Moon is Pareto non-dominated in **every** cell - the sole consensus analog - and is the *decisive* recommendation for the volcanic plains under three of four lenses, while the crater floors tie more often (no clean crater-floor analog among the three). The engine points confidently where the morphology agrees and abstains where it does not. See `REAL-EXAMPLE.md`.

## 5. Validation

Correctness is pinned by a scientific-correctness suite of 51 oracles that drive the real engine on synthetic terrains with known answers: self-distance is zero; slope/relief/ruggedness are invariant to a constant offset and to 90 deg rotation; roughness discrimination is monotone; a planted close twin is recovered over an impostor; a water tile is BLOCKED; openness is 90 deg on a plane and exceeds 90 deg at a peak; geomorphons label a cone apex a peak and a basin centre a pit; a sinusoid's spectral peak lands at its wavelength; white noise is spectrally flat (Parseval holds); a ridged surface reports the correct fabric orientation. The real-data run adds an external check: the method re-identifies a documented basalt analog on the core axes without being told to.

## 6. Limitations and transfer gaps (stated, not hidden)

TerraTrace compares *map-scale morphology of sampled DEMs* and nothing else. It says nothing about composition, rock-scale obstacles below the pixel, trafficability, or mission suitability; those bounds are reproduced verbatim in every report. Two concrete limits of the worked example: the Mars window (~10 deg) and the Earth windows (~0.4 deg) differ in footprint by ~15x, so the screen mixes regional Mars structure with local Earth structure (a transfer gap the ledger flags; the fix is comparable-footprint AOIs); and the independent-pixel bootstrap is optimistic relative to the spatial block bootstrap on autocorrelated terrain. Deferred work: DEM co-registration and vertical-datum error propagation end-to-end into the decision, and a multi-pair "analog atlas."

## 7. Reproducibility

```
pip install -r terratrace/requirements.txt      # core: NumPy only
python terratrace/tests/run_tests.py             # 686 offline tests
python terratrace/scripts/run_validation.py      # 51 correctness oracles
# real five-world study, one command (needs a free OpenTopography key):
python terratrace/scripts/run_real_analog_study.py --config config/real_multibody.example.yml
```

CI runs the suite and the validation oracles on Python 3.10/3.11/3.12. Large DEMs and generated bundles are git-ignored; everything else regenerates deterministically.

## References


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*Morphology screening and resampling statistics only - not a traversability, geologic-equivalence, site-access, or mission-readiness claim.*